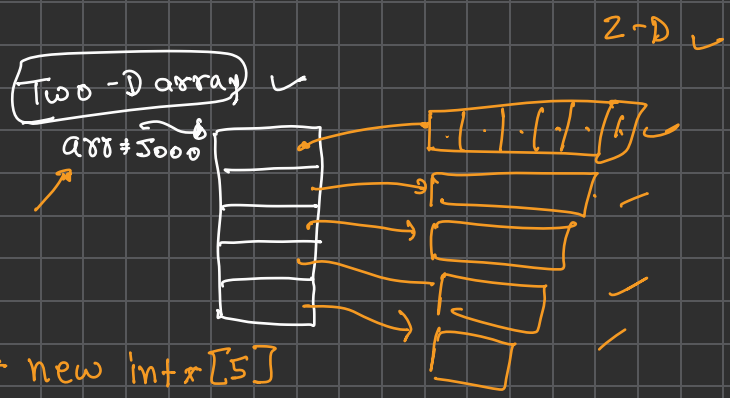
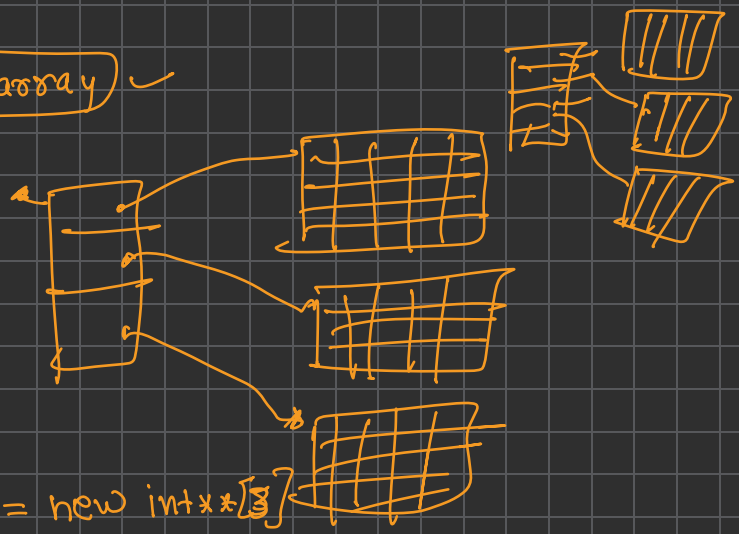



```
int **arr = new int*[5]

For(int i=0 ;i<5 ;i++)
  {   arr[i] = new int[5];
  }
```

3 D array

arr



```
int *** arr = new int**[3];
```

Recursion:

- → A function which call itself again;
- → finally stop calling itself.

→ Bachon ko dekha hoga

if placed

- → wake-up
- → college jaate hai
- → Padai karte
- → Sojata

Eat
Sleep
Gym
Repeat

```
function Schedule()
{
    if (placed)
        return;

    ① wakeup
    ② Go to college
    ③ Padai
    ④ sleep

    schedule()
}
```

```
Scedule();


function Gym()
{       if (dead)
            return;

    → Go to Gym
    → Eat clean
    → Sleep
    Gym();
}
```

① Instagram: whatsapp: → Besties

30 days left for birth
29 day left for birth
28 days . .
.
.
.
1 day left
Happy birthday!

→ Print(5) = 5, Print(4)

Pint(4) = 4, Print(3)

Print(3) = 3, Print(2)

Print(2) = 2, Print(1)

Print(1) = 1, Print(0)

$\boxed{\text{Print(0) = return}}$

n to 1

5

4

3

2

1

0 stop

function

Print(n) = n, Print(n-1);

print(0) = return;

10 to 1

⇓

end

PrintEven(0) ⇒ return;

PrintEven(2) = 2, PrintEven(0)

Print(Even(4) = 4, PrintEven(2)

Print Even (8) = 8, Print(Even(6)

Print Even (10) = 10, print Even (8)

10

8

6

4

2

0 × return

PrintEven (n) ⇒ n, printEven(n-2)

1 to n

Print(x) = (x==n+1) return;

Print(5) = 5, print(6)

Print(4) = 4, Print(5)

Print(3) = 3, Print(4)

Print(2) = 2, Print(3)

Print(1) ⇒ 1, print(2)

Print(1)

1 to 5 nom

1
2
3
4
5
6 X stop

Print(x, n) = (x==n+1) return;

Print(5, n) ⇒ 5, print(6, n)

print(4, n) = 4, print(5, n)

⋮

[Print(1, n) = 1, print(2, n)]

1
2
3
⋮
n
n+1 → stop

(1) Print(x, n) = x, print(x+1, n);

$\boxed{5+0!}$

SumNumber(5) = 5 + SumNumber(4)

Sum Number(4) = 4 + Sum Number(3)

Sum Number(3) = 3 + SumNumber(2)

Sumber(2) = 2 + SumNumber(1)

SumNumber(1) = 1 + SumNumber(0)

5
4
3
2
1
0 ← $\boxed{\text{return}}$

SumNumber = 0;

return

SumNumber(n) = n + SumNumber(n-1)

SumNumber(0) = 0, return

Base condition

5! = 5 x 4!
4! = 4 x 3!
3! = 3 x 2!
2! = 2 x 1!
1! = 1

Factorial(5) ⇒ 5 * factorial(4)

Factorial(n) = n * Factorial(n-1)

$5 \times 4 \times 3 \times 2 \times 1$

$10! = 10 * 9!$

$9! = 9 * 8!$

$fact(n) = n * fact(n-1)$

$fact(1) = 1$